

ECON 351 - Assumptions & Preferences Pro

Screenshots from homework, sample exam, and mock midterm

Sources checked:

- Homework 1 Ch.4 Ex 2: 3 assumption questions (completeness, transitivity)
- Sample Questions Ch.4: Q5-Q8, Q30-Q31 (IC slope, substitutes, complements)
- Mock Midterm 1: Q22 (Cobb-Douglas same IC)
- Homework 2 Ch.9: No assumptions/preferences questions
- Sample Questions Ch.9: No assumptions/preferences questions
- Practice Tests 1-3: 2 assumptions/preferences questions each

9 problem sets in this PDF.

Assumptions & Preferences - Formula Sheet

Completeness: consumer can always rank any two bundles (or be indifferent)

Transitivity: if $A > B$ and $B > C$, then $A > C$

Cannot decide between bundles -> violates completeness

Indifference is OK -> does NOT violate completeness

IC slope (absolute value) = $MRS = MU_x / MU_y$

Perfect substitutes: straight-line ICs, $u = aX + bY$

Perfect complements: L-shaped ICs, fixed proportions

Diminishing MRS: standard convex ICs bow toward origin

Problem 1: P-1

Homework 1 Ch.4 - Exercise 2 (2.1, 2.2, 2.3)

Jackie movie theater scenarios - completeness and transitivity.

Exercise 2 - Preferences' Assumptions

2.1 Jackie was at the movie theater last night. The two movies available were "Knives out" and "Top Gun Maverick." She didn't care which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

2.2 Again, Jackie goes to the movies. The two movies available are "Avatar" and "The Avengers," and she says she cannot decide which one to see. Does this imply a violation of any assumption on preferences? If so, which one?

2.3 On a different day, there were three movies available: "Parasite," "Indiana Jones," and "Jurassic Park." Jackie said she rather watch Parasite over Indiana Jones and Indiana Jones over Jurassic Park. But she would also prefer watching Jurassic Park over Parasite. Does this imply a violation of any assumption on preferences (beyond displaying poor taste)? If so, which one?

Problem 2: P-2

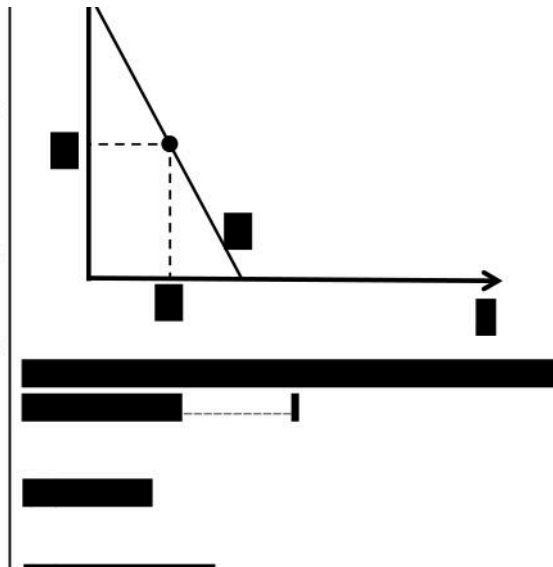
Sample Questions Ch.4 - Question 5 | Practice Test 3

What does the slope of an IC tell you?

Question 5:

By observing the slope of an indifference curve, we can infer

- (A) that preferences are complete.
- (B) the marginal utility of the good on the vertical axis.
- (C) the marginal rate of substitution of one good for another good.
- (D) the ratio of market prices.
- (E) that preferences are transitive.



Problem 3: P-3

Sample Questions Ch.4 - Question 6 | Practice Test 2

Magnitude of IC slope = MRS (two-column page).

Question 6:

The magnitude of the slope of an indifference curve is:

- (A) called the marginal rate of substitution.
- (B) equal to the ratio of the total utility of the goods.
- (C) always equal to the ratio of the prices of the goods.

(D) all of the above.

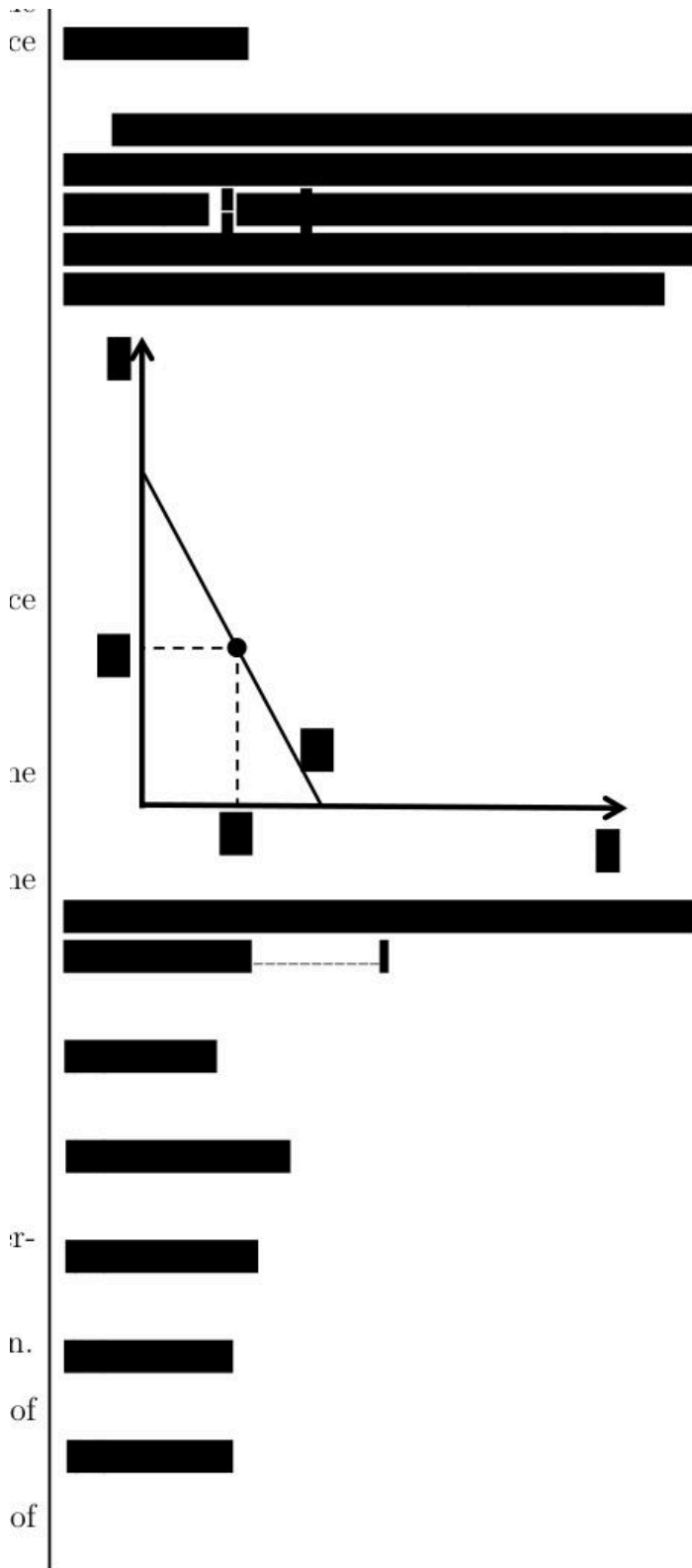
(E) A and C only.

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Problem 4: P-4

Sample Questions Ch.4 - Question 7

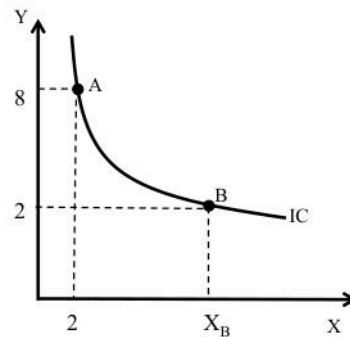
Inka: linear utility $u = (3/4)X + (2/7)Y$ - perfect substitutes.



Problem 5: P-5

Sample Questions Ch.4 - Question 8 | Practice Test 1

Serena: straight-line IC slope $-7/5 \rightarrow u = X/5 + Y/7$.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Question 10:

Assume that apples are measured on the horizontal axis and oranges on the vertical axis. If the price of apples increases from \$3 to \$6, and the price of oranges increases from \$2 to \$4, then the budget line will:

- (A) become flatter.
- (B) become steeper.
- (C) shift inward (parallel shift).
- (D) shift outward (parallel shift).

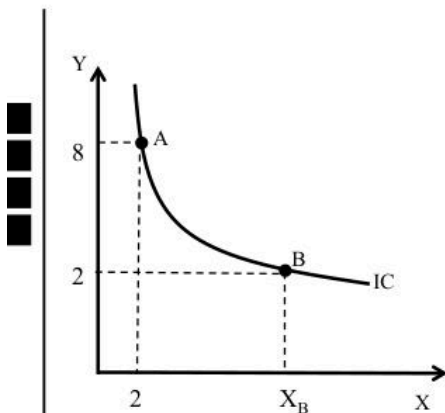
Problem 6: P-6

Sample Questions Ch.4 - Question 9

Angelique: Cobb-Douglas baskets on same IC (two-column page).

Question 9:

Angelique's preferences over goods X and Y can be represented by the following utility function: $u(X, Y) = X^2Y^3$. The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



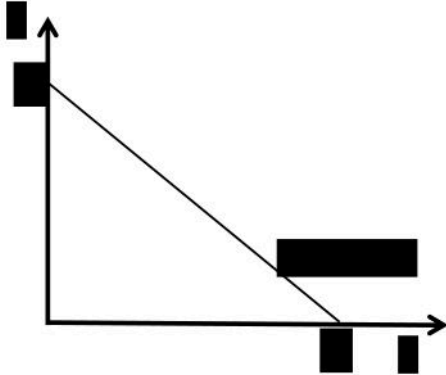
What do we know about X_B ?

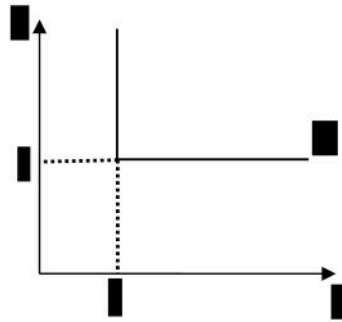
- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Problem 7: P-7

Sample Questions Ch.4 - Question 30

Tony: perfect complements $1/4$ X with $1/6$ Y.

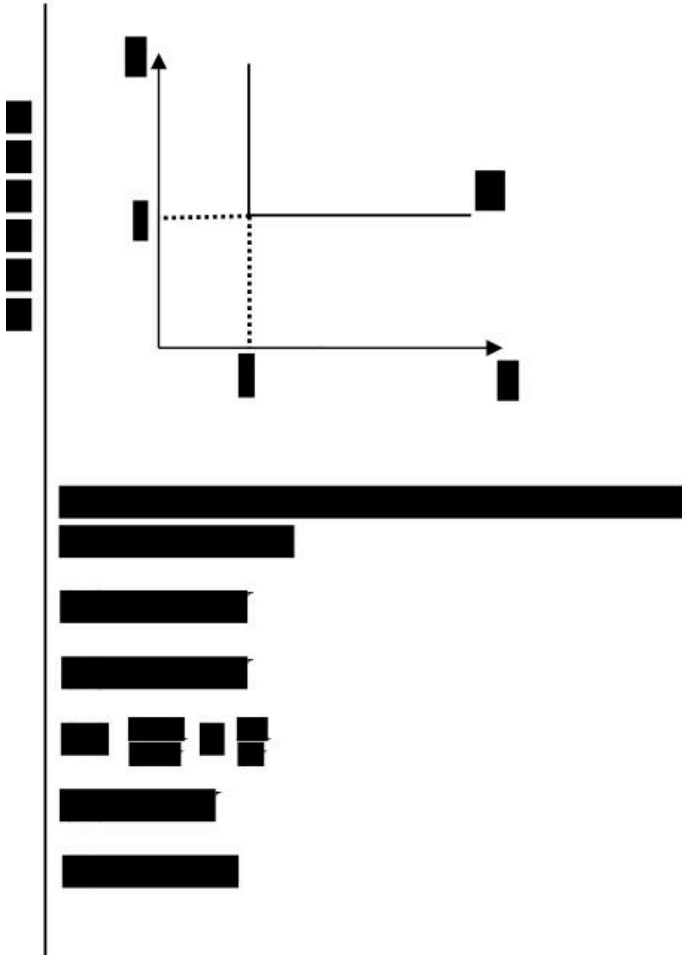




Problem 8: P-8

Sample Questions Ch.4 - Question 31

Jimmy: L-shaped IC - optimal mix $9X = 4Y$ (two-column page).



Problem 9: P-9

Mock Midterm 1 - Question 22

Cobb-Douglas $u = X^6 Y^3$ - find X_B on same IC as basket A.

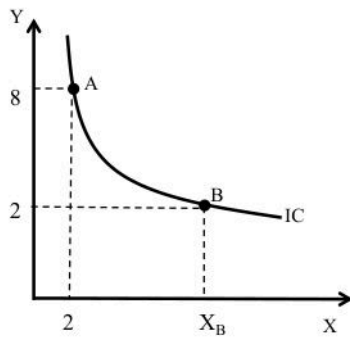
Question 22:

[Show your work]

A consumer has preferences over goods X and Y that can be represented by the following utility function:

$$u(X, Y) = X^6 Y^3.$$

The graph below represents two baskets, A and B, on the same indifference curve of this consumer.



What do we know about X_B ?

- (A) $X_B = 4$
- (B) $X_B = 3$
- (C) $X_B = 18$
- (D) $X_B = 16$
- (E) $X_B = 8$

Answer Key

P-1 (2.1):

No violation. Indifference is allowed under completeness.

P-1 (2.2):

Yes - violates completeness (cannot rank bundles).

P-1 (2.3):

Yes - violates transitivity.

P-2:

(C) the marginal rate of substitution.

P-3:

(A) called the marginal rate of substitution.

P-4:

(B) $Y = 17.333$

P-5:

(D) $u(X,Y) = X/5 + Y/7$

P-6:

(D) $X_B = 16$

P-7:

(D) $6X = 4Y$

P-8:

(A) $9X = 4Y$

P-9:

(D) $X_B = 16$